Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/30/2006
PI Name:	Ingber, Donald M.D., Ph.D.		
Project Title:	Mechanotransduction Through Integrins		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	02115	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 6/30/2008 per A. Chu/ARC (6/07) 8/06--funding reinstated thru 6/30/007, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:	Over the past year, we hired Dr. Jessica Tytell, an expert in molecular cell biology to assist us in these studies.		
COI Name (Institution):			
Grant/Contract No.:	NNA04CC96G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The specific aims of this project are 1. To compare the effects of mechanical stresses of different intensity, frequency and direction on focal adhesion assembly, cell strengthening and G _protein activation. 2. To analyze the Rho signaling mechanism responsible for cell strengthening behavior. 3. To identify the molecules that mediate transmembrane mechanochemical coupling between integrins and G proteins within the focal adhesion. Over the past year, we showed that intracellular microtubules do bear large-scale compressive loads from a variety of physiological forces, but their buckling wavelength is reduced significantly because of mechanical coupling to the surrounding elastic cytoskeleton. We quantitatively explained this behavior, and showed that this coupling dramatically increases the compressive forces that microtubules can sustain, suggesting they can make a more significant structural contribution to the mechanical behavior of the cell than previously thought possible.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Elucidation of the mechanism by which integrins mediate cellular mechanotransduction should help us to understand the molecular basis of the cellular response to gravity as well as many other forms of mechanosensation and tissue regulation. Through the proposed research program, we hope to identify specific molecular mediators of mechanochemical transduction as well as novel mechanical signaling pathways in mammalian cells. These molecules may serve as potential targets for future modes of therapeutic intervention in diseases, such as atherosclerosis, hypertension, and osteoporosis, which are characterized by abnormal force transduction by cells. Further insight into mechanobiology also may impact on design of artificial tissues and medical devices for tissue repair and transplantation.
Task Progress:	The formation of focal adhesions that mediate alterations of cell shape and movement is controlled by a mechanochemical mechanism in which cytoskeletal tensional forces drive changes in molecular assembly; however, little is known about the molecular biophysical basis of this response. Over the past year, we described a method to measure the unbinding rate constant of individual GFP-labeled focal adhesion molecules in living cells by modifying the fluorescence recovery after photobleaching (FRAP) technique and combining it with mathematical modeling. Using this method, we show that decreasing cellular traction forces on focal adhesions by three different techniques – chemical inhibition of cytoskeletal tension generation, laser incision of an associated actin stress fiber, or use of compliant extracellular matrices – increases the of the focal adhesion protein zyxin. In contrast, the of another adhesion protein, vinculin, remains unchanged after tension dissipation. Mathematical models also demonstrate that these force-dependent increases in zyxin's that occur over seconds are sufficient to quantitatively predict large-scale focal adhesion disassembly that occurs physiologically over many minutes. These findings demonstrate that the molecular binding kinetics of some, but not all, focal adhesion proteins are sensitive to mechanical force, and suggest that force-dependent changes in this biophysical parameter may govern the supramolecular events that underlie focal adhesion remodeling in living cells. The ability of cells to sense local variations in mechanical forces transmitted across extracellular matrix adhesions and respond by strengthening their integrin-cytoskeleton linkages within focal adhesions is critical for control of cell shape, growth and motility. To better understand how cells adapt to changes in mechanical forces exerted on their extracellular matrix adhesions, we applied tensional forces to magnetic microbeads bound to cell surface integrin receptors using magnetic pulling cytometry and measured dynamic changes in bead displacement with sub-micrometer resolution using optical microscopy. These studies revealed that cells exhibit four distinct types of mechanical responses: 1) an immediate viscoelastic response to a brief 3 second force pulse that resulted in almost complete elastic recovery, 2) early adaptive behavior characterized by pulse-to-pulse attenuation in bead displacement in response to an oscillatory force regimen composed of multiple brief (3 second) force pulses separated by short (4 second) relaxation periods, 3) later adaptive cell stiffening in response to sustained (> 15 second) static stresses, and 4) a large-scale repositioning response in which beads exposed to prolonged (> 1 minute) static stresses were retracted back by the cell against the applied force. Importantly, these forms of cellular adaptation differed biochemically. Whereas the immediate viscoelastic response and early adaptation behavior were affected by dissipating cytoskeletal prestress (isometric tension) with inhibitors of either myosin ATPase or Rho-associated kinase (ROCK), the later adaptive response to longer (> 15 second) stresses was not sensitive to either of these inhibitors. The large-scale repositioning of beads in response to prolonged (> 1 minute) stress was prevented by inhibiting myosin-based tension generation through interference with Rho signaling, similar to the case of the immediate and early responses, but in addition it was prevented by blocking mechanosensitive ion channels using gadolinium or by inhibiting tyrosine kinases using PP2. All of these adaptive responses were suppressed by cooling the cells to 48C to slow biochemical remodeling events, and the large-scale repositioning response required that integrins be chemically activated through occupancy of the RGD binding site. These results demonstrate that cells use multiple distinct mechanisms to sense and respond to static and dynamic changes in the physical properties of the extracellular matrix that alter the level of mechanical stress applied to cell surface integrin receptors. Over the last year, we also completed studies initiated in prior NASA grant funding periods that were designed to test the cellular tensegrity hypothesis by determining whether microtubules function as compression struts in living cells. Key to this theory of cell structure and mechanotransduction is that cytoskeletal microtubules influence cell shape and mechanics based on their ability to resist large-scale compressive forces exerted by the surrounding contractile cytoskeleton. Consistent with this, cytoplasmic microtubules are often highly curved and appear buckled due to compressive loads. However, results of in vitro studies suggest that microtubules should buckle at much larger length scales, withstanding only exceedingly small compressive forces. This discrepancy calls into question the structural role of microtubules, and highlights our lack of quantitative knowledge of the magnitude of the forces they experience and can withstand in living cells. Over the past year, we showed that intracellular microtubules do bear large-scale compressive loads from a variety of physiological forces, but their buckling wavelength is reduced significantly because of mechanical coupling to the surrounding elastic cytoskeleton. We quantitatively explained this behavior, and showed that this coupling dramatically increases the compressive forces that microtubules can sustain, suggesting they can make a more significant structural contribution to the mechanical behavior of the cell than previously thought possible. Finally, our research on how living cells sense and response to mechanical forces, such as gravity, which has been funded continuously by NASA over the past 20 years, has culminated into new insight into the multi-scale nature of this process. In the past, analysis of cellular mechanotransduction has focused on identification of critical mechanosensitive molecules and cellular components. Stretch-activated ion channels, caveolae, integrins, cadherins, growth factor receptors, myosin motors, cytoskeletal filaments, nuclei, extracellular matrix and numerous other structures and signaling molecules all have been shown to contribute to the mechanotransduction response. However, little is known about how these different molecules function within the structural context of living cells, tissues and organs to produce

	the orchestrated cellular behaviors that are required for mechanosensation, embryogenesis and physiological control. As we described in an article published over the past year, recent work from a wide range of fields reveals that organ, tissue and cell anatomy are as important for mechanotransduction as individual mechanosensitive proteins, and that our bodies use structural hierarchies (systems within systems) composed of interconnected networks that span from the macroscale to the nanoscale in order to focus stresses on specific mechanotransducer molecules. The presence of isometric tension (prestress) at all levels of these multiscale tensegrity-based networks ensures that various molecular-scale mechanochemical transduction mechanisms proceed simultaneously and produce a concerted response. Future research in this area will therefore require analysis, understanding and modeling of tensionally-integrated (tensegrity) systems of mechanochemical control.
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